

Python Datastructure

Task 1

-list -tuple -set -dict -range

```
In [223...] 1 = []  
1
```

```
Out[223...] []
```

```
In [224...] len(1)
```

```
Out[224...] 0
```

```
In [225...] l1 = [10,20,30]  
l1
```

```
Out[225...] [10, 20, 30]
```

```
In [226...] len(l1)
```

```
Out[226...] 3
```

```
In [227...] l1.append(40)  
l1.append(50)  
l1.append(60)  
l1.append(10)
```

```
In [228...] l1.append(70,80)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[228], line 1  
----> 1 l1.append(70,80)  
  
TypeError: list.append() takes exactly one argument (2 given)
```

```
In [229...] l1.append(['a','b','c'])
```

```
In [230...] l2 = [34,5.3,'ajim',(1+2j),True]
```

```
In [231...] print(l)  
print(l1)  
print(len(l2))
```

```
[]  
[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]  
5
```

```
In [232... print(id(l))
            print(id(l1))
            print(id(l2))
```

```
2042089036544
2042089054144
2042089230464
```

```
In [233... print(type(l))
            print(type(l1))
            print(type(l2))
```

```
<class 'list'>
<class 'list'>
<class 'list'>
```

```
In [234... l1
```

```
Out[234... [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [235... # List membership
            10 in l1
```

```
Out[235... True
```

```
In [236... ['a','b','c'] in l1
```

```
Out[236... True
```

```
In [237... l1
```

```
Out[237... [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [238... 100 in l1
```

```
Out[238... False
```

```
In [239... l2
```

```
Out[239... [34, 5.3, 'ajim', (1+2j), True]
```

```
In [240... l3 = l2.copy()
```

```
In [241... l3
```

```
Out[241... [34, 5.3, 'ajim', (1+2j), True]
```

```
In [242... print(id(l2))
            print(id(l3))
```

```
2042089230464
2042087823552
```

```
In [31]: print(l)
         print(l1)
         print(l2)
         print(l3)

[]
[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
[34, 5.3, 'ajim', (1+2j), True]
[34, 5.3, 'ajim', (1+2j), True]
```

```
In [243... 13
```

```
Out[243... [34, 5.3, 'ajim', (1+2j), True]
```

```
In [244... len(l3)
```

```
Out[244... 5
```

```
In [245... l3.clear()
```

```
In [246... 13
```

```
Out[246... []
```

```
In [247... len(l3)
```

```
Out[247... 0
```

```
In [248... del l3    # we delete the function l3 permanataly
```

```
In [249... print(l)
         print(l1)
         print(l2)

[]
[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
[34, 5.3, 'ajim', (1+2j), True]
```

```
In [250... l2.count(5.3)    # count is repetation ,5.3 is 1 time repeated in l2 function
```

```
Out[250... 1
```

```
In [251... l1.count(10)    # here 10 is repeated 2 times in l1 function
```

```
Out[251... 2
```

```
In [252... l1
```

```
Out[252... [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [253... l1[:]
```

```
Out[253... [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

In [254... `l1[-1]`

Out[254... `['a', 'b', 'c']`

In [255... `l1`

Out[255... `[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]`

In [256... `l1[100]`

```
-----  
IndexError                                Traceback (most recent call last)  
Cell In[256], line 1  
----> 1 l1[100]  
IndexError: list index out of range
```

In [257... `l1`

Out[257... `[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]`

In [258... `l1[4:]`

Out[258... `[50, 60, 10, ['a', 'b', 'c']]`

In [259... `l1`

Out[259... `[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]`

In [260... `l1[3:-1]`

Out[260... `[40, 50, 60, 10]`

In [261... `l1`

Out[261... `[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]`

In [262... `l1[0:10:3]`

Out[262... `[10, 40, 10]`

In [263... `l1`

Out[263... `[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]`

In [264... `l1[::3]`

Out[264... `[10, 40, 10]`

In [265... `l1`

Out[265... `[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]`

In [266... `l1[::1]`

Out[266... `[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]`

In [267... `l1`

Out[267... `[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]`

In [268... `l1[::-1]`

Out[268... `[['a', 'b', 'c'], 10, 60, 50, 40, 30, 20, 10]`

In [269... `l1`

Out[269... `[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]`

In [270... `l1[::-2]`

Out[270... `[['a', 'b', 'c'], 60, 40, 20]`

In [271... `l1`

Out[271... `[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]`

In [272... `l1[::-3]`

Out[272... `[['a', 'b', 'c'], 50, 20]`

In [273... `l1`

Out[273... `[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]`

In [316... `print(l)
print(l1)
print(l2)`

`[True, (1+2j), 'ajim', 5.3, 34, True, (1+2j), 'ajim', 5.3, 34, True, (1+2j), 'ajim',
34, ['a', 'b', 'c'], 10, 60, 50, 30, 20, 10]
[100, 20, 30, 40, 50, 60, 10, 'Azim Mujawar', 10]
[34, 5.3, 25, 'ajim', (1+2j)]`

In [275... `l1`

Out[275... `[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]`

In [276... `l1[0]`

Out[276... `10`

In [277... `l2[0]`

Out[277... `34`

In [278... 11

Out[278... [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

```
In [317... print(l)
print(l1)
print(l2)
```

```
[True, (1+2j), 'ajim', 5.3, 34, True, (1+2j), 'ajim', 5.3, 34, True, (1+2j), 'ajim',
34, ['a', 'b', 'c'], 10, 60, 50, 30, 20, 10]
[100, 20, 30, 40, 50, 60, 10, 'Azim Mujawar', 10]
[34, 5.3, 25, 'ajim', (1+2j)]
```

```
In [280... l.extend(l1)
```

In [281... 1

Out[281... [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

In [282... 11

Out[282... [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

```
In [283... 11[0]=100
```

In [284... 11

Out[284... [100, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

```
In [285... 11[-1]='Azim Mujawar'
```

```
In [329... print(l)
print(l1)
print(l2)
```

```
[True, (1+2j), 'ajim', 5.3, 34, True, (1+2j), 'ajim', 5.3, 34, True, (1+2j), 'ajim',
34, ['a', 'b', 'c'], 10, 60, 50, 30, 20, 10]
[10, 20, 30, 40, 50, 60, 100, 'Azim Mujawar', 10]
[5.3, 25, 34, 'ajim', (1+2j)]
```

```
In [330... l.extend(l2)
```

In [295... 12

Out[295... [34, 5.3, 'ajim', (1+2j), True]

```
In [296... 1[0]
```

Out[296... 10

In [297... 12

Out[297... [34, 5.3, 'ajim', (1+2j), True]

In [328... `l1.index(l1)`

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[328], line 1  
----> 1 l1.index(l1)  
ValueError: [10, 20, 30, 40, 50, 60, 100, 'Azim Mujawar', 10] is not in list
```

In [299... `l2`

Out[299... `[34, 5.3, 'ajim', (1+2j), True]`

In [300... `l1.append(10)`

In [301... `l2`

Out[301... `[34, 5.3, 'ajim', (1+2j), True]`

In [302... `l2.insert(2,25)`

In [303... `l2`

Out[303... `[34, 5.3, 25, 'ajim', (1+2j), True]`

In [304... `l2.pop()`

Out[304... `True`

In [305... `l2`

Out[305... `[34, 5.3, 25, 'ajim', (1+2j)]`

In [306... `l[3]`

Out[306... `40`

In [307... `l2`

Out[307... `[34, 5.3, 25, 'ajim', (1+2j)]`

In [308... `l2[-2]`

Out[308... `'ajim'`

In [309... `l.pop(3)`

Out[309... `40`

In [311... `l.remove(5.3)`

In [312... `l1`

Out[312... [100, 20, 30, 40, 50, 60, 10, 'Azim Mujawar', 10]

In [313... `l.reverse()`

In [314... `l1`

Out[314... [100, 20, 30, 40, 50, 60, 10, 'Azim Mujawar', 10]

In [320... `print(l)`
`print(l1)`
`print(l2)`

[True, (1+2j), 'ajim', 5.3, 34, True, (1+2j), 'ajim', 5.3, 34, True, (1+2j), 'ajim',
 34, ['a', 'b', 'c'], 10, 60, 50, 30, 20, 10]
 [100, 20, 30, 40, 50, 60, 10, 'Azim Mujawar', 10]
 [34, 5.3, 25, 'ajim', (1+2j)]

In []:

In [332... `l1`

Out[332... [10, 20, 30, 40, 50, 60, 100, 'Azim Mujawar', 10]

In [333... `l1.sort()`

```
-----
TypeError                                Traceback (most recent call last)
Cell In[333], line 1
----> 1 l1.sort()

TypeError: '<' not supported between instances of 'str' and 'int'
```

In [334... `l1`

Out[334... [10, 20, 30, 40, 50, 60, 100, 'Azim Mujawar', 10]

In [335... `l1 = [3,1,2]`

In [336... `l1`

Out[336... [3, 1, 2]

In [337... `l1.sort()` *# parameter tuning*

In [338... `l1` *#....ascending order*

Out[338... [1, 2, 3]

In [339... `l1.sort(reverse=True)` *# Hyperparameter tuning*
`l1` *#.....desending order*

Out[339... [3, 2, 1]

In [341... 12

Out[341... [5.3, 25, 34, 'ajim', (1+2j)]

In [343... 12[3]

Out[343... 'ajim'

In [344... 12

Out[344... [5.3, 25, 34, 'ajim', (1+2j)]

In [345... 12[3][0]

Out[345... 'a'

In [346...
 print(12[3][0])
 print(12[3][1])
 print(12[3][2])
 print(12[3][3])

a
 j
 i
 m

In [351... 12

Out[351... [5.3, 25, 34, 'ajim', (1+2j)]

In [352...
 for i in 12:
 print(i)

5.3
 25
 34
 ajim
 (1+2j)

In [359...
 for i in enumerate(12):
 print(i)

(0, 5.3)
 (1, 25)
 (2, 34)
 (3, 'ajim')
 (4, (1+2j))

In [360... 12

Out[360... [5.3, 25, 34, 'ajim', (1+2j)]

In [361... all(12)

Out[361... True

In [362... `any(l2)`

Out[362... True

In [363... `l2.append(0)`

In [364... `l2`

Out[364... `[5.3, 25, 34, 'ajim', (1+2j), 0]`

In [365... `all(l2)`

Out[365... False

In [366... `any(l2)`

Out[366... True

In [367... `l3 = []`

In [368... `all(l3)`

Out[368... True

In [369... `any(l3)`

Out[369... False

In []: `#`